CUSTOMER CHURN PREDICTION

EXECUTIVE SUMMARY

1. Background

Customer churn is a major challenge for banks of all sizes. According to a recent study by Bain & Company, the average cost of acquiring a new customer is five times the cost of retaining an existing one. This means that it is critical for banks to identify and address the factors that are leading to customer churn.

Customer churn impaired is the process of losing customers who are at high risk of churning. This can happen for a variety of reasons, such as dissatisfaction with the product or service, poor customer service, or competitive offerings. Customer churn impaired can have a significant impact on a company's bottom line, as it costs more to acquire a new customer than to retain an existing one.

churn impaired? impaired in banking is no payment but like customer no satisfy so that they will not pay???

There are several key trends and drivers of customer churn impaired. One trend is the rise of the digital economy. Customers have more choices than ever before, and they are more likely to switch to a competitor if they are not satisfied with their current provider. Another trend is the increasing importance of customer experience. Customers expect a personalized and seamless experience across all touchpoints. If a company fails to meet these expectations, customers are more likely to churn.

this means that customer do early repayment and fully settle, there is no term on impaired account if customer fully settle

Understanding the key trends and drivers of customer churn impaired is essential for developing effective strategies to reduce churn. By identifying and addressing the factors that are driving customers away, companies can improve customer satisfaction and retention.

2. Problem statement

dont you think the number of customers reduced due to fully settle account?

Customer churn is a major problem for our business. In the past year, our number of customers has decreased by around **30%**, gross financing assets have decreased by **17%** and cumulative profit payment slight decrease by **18%** while our provisions slight has a decrease but maintain around the same from **December 2020** until **September 2023**. This raises the concern that the bank's customer becomes impaired. This will also result in a significant loss of revenue and profitability. We need to identify and address the root causes of customer churn to reduce the number of customers who are leaving our business.

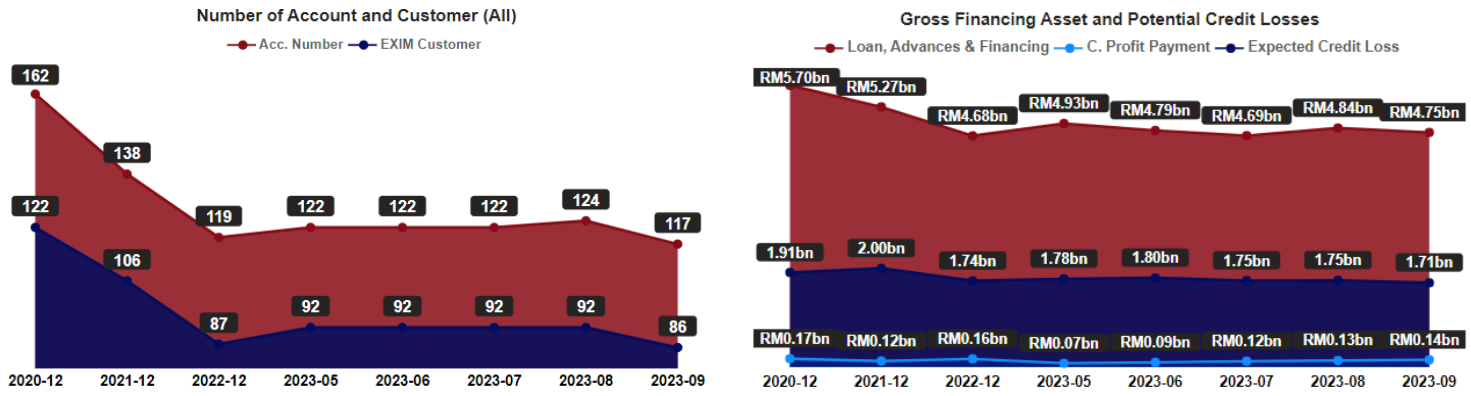


Figure 1.0

The figure above shows the movement of EXIM Bank losing customers with its cumulative profit faster than the bank can bring them on board (Refer Account Migration Section). Meanwhile, another concern to investigate is that the Gross Financing Asset

is decreasing, but the provision amount at one point is increasing. It shows the quality of the existing customer is not good either.

3. Impact

customer attrition? meaning?

Our Bank is significantly being impacted by customer attrition. Due to churn from **December 2020** to **September 2023**, we have lost **RM 0.95 billion** in our gross financing asset and **RM0.03 billion** in cumulative profit payment throughout the year. Churn is a problem that will continue to harm the business if it's not treated properly.

4. Detailed

Business Performance

EXIM Bank's primary line of business and profit is non-trade financing. Throughout the year, both trade and non-trade customers, as well as its financing asset's client base, are steadily diminishing while accrued profit and profit payments sluggishly increase throughout the year:

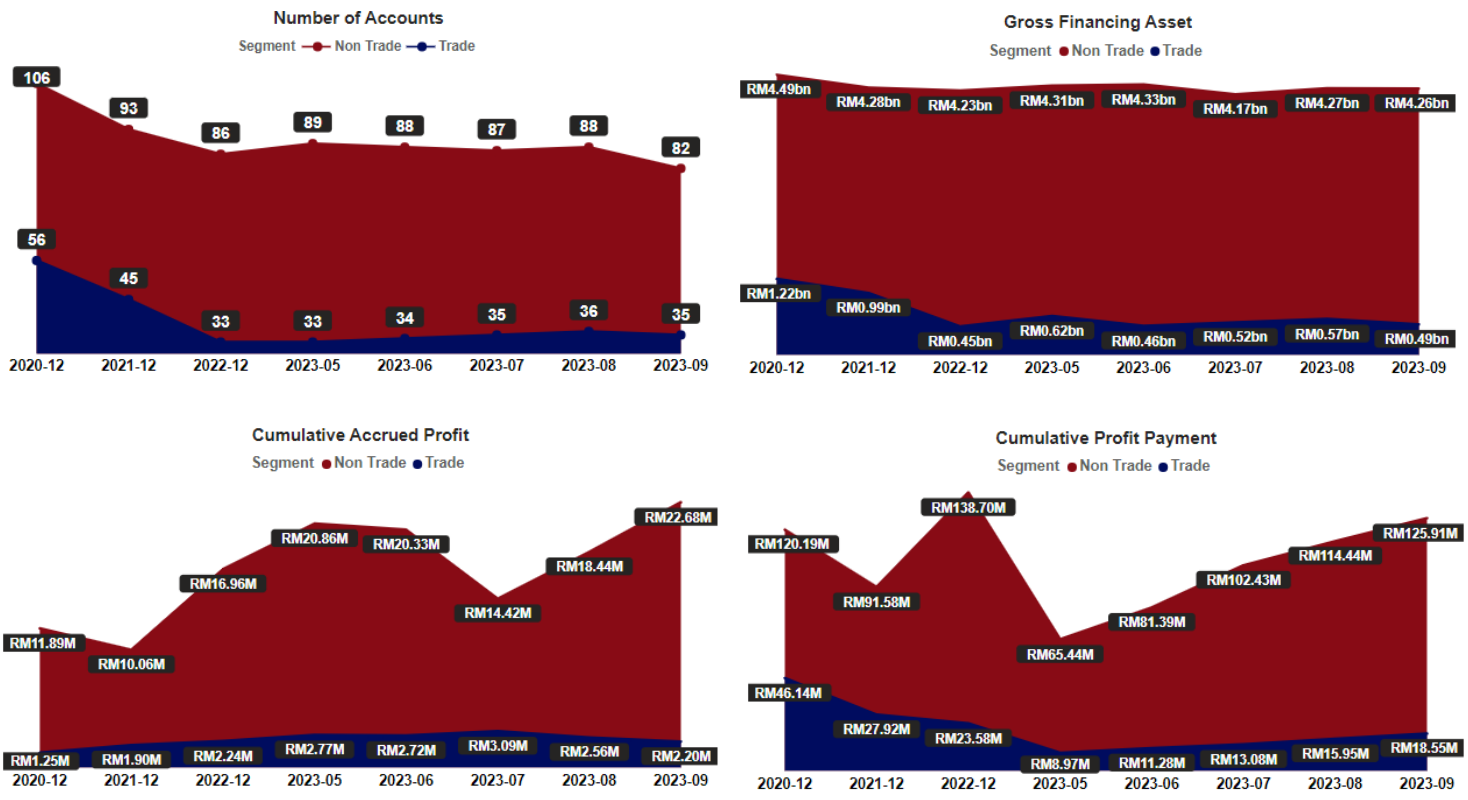
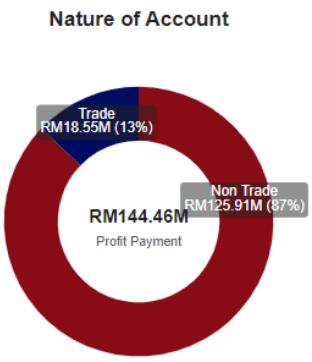


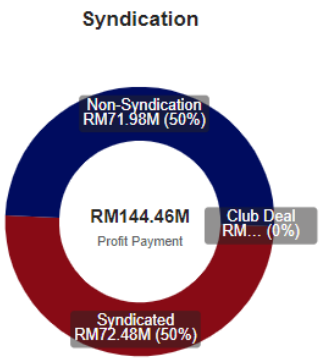
Figure 2.0

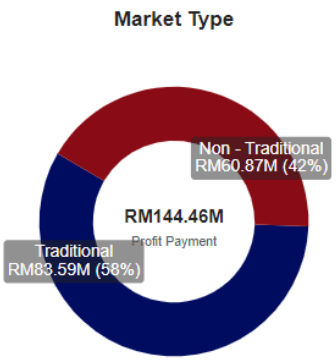
Customer Portfolio: September 2023

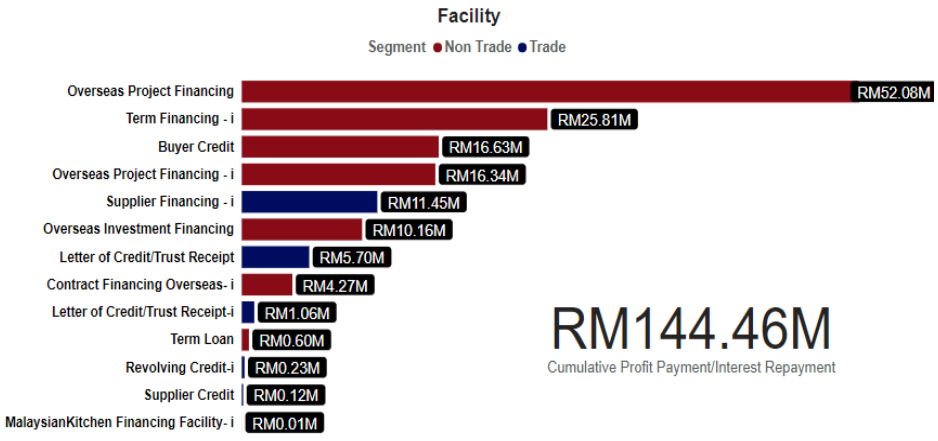
The chart below displays **cumulative profit payment** from EXIM clients as of the most recent position. Most profit generated are generally from non-trade, syndicated, and traditional markets, as well as term financing-i and overseas project financing

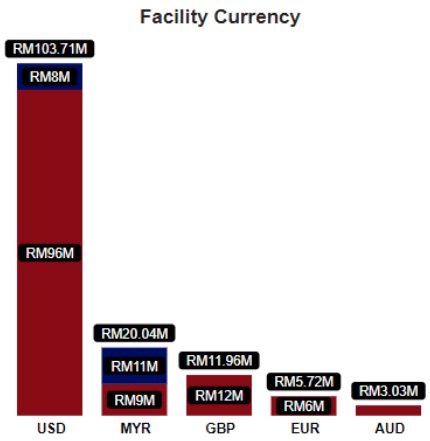
facilities. With currencies of USD, Malaysia and Brazil generated the most profit from EXIM Bank customers. Aside from that, Active clients, non-SME entities in the and manufacturing and business industries as making profit as well.

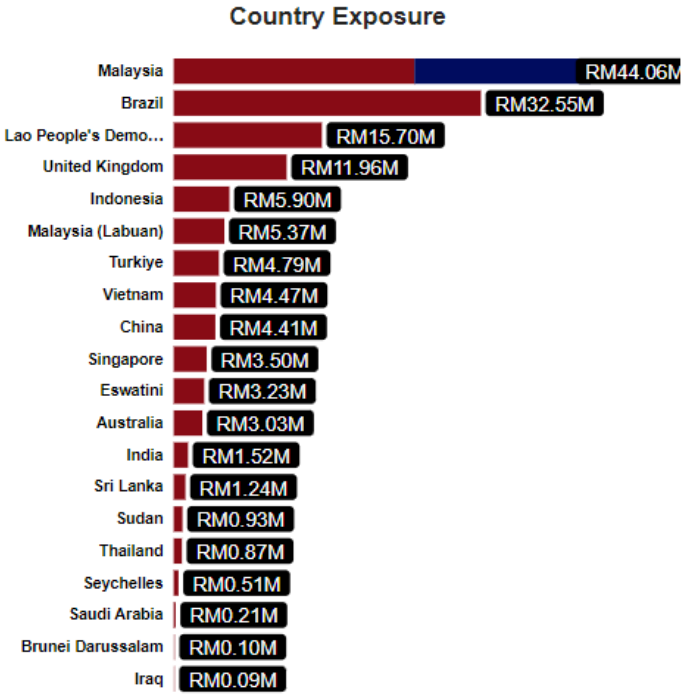


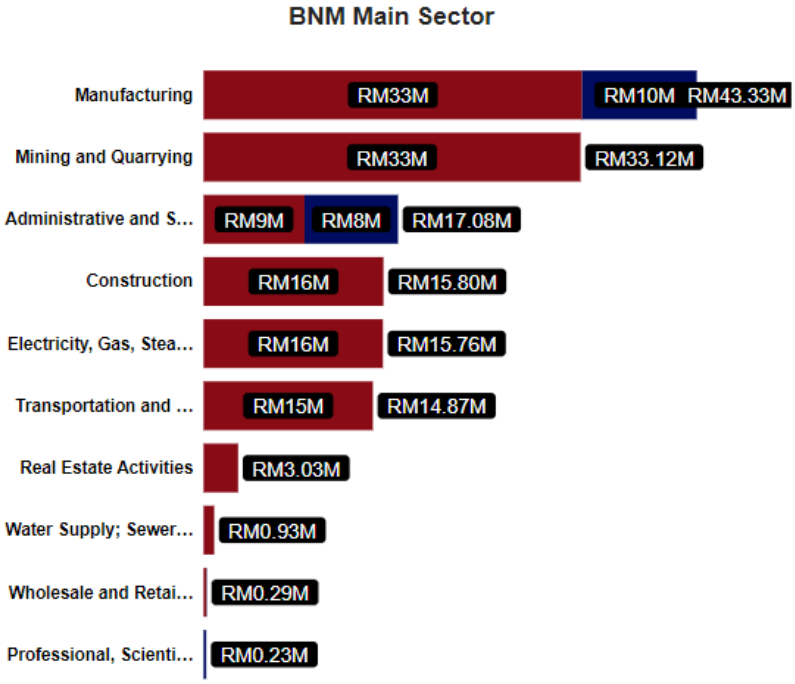












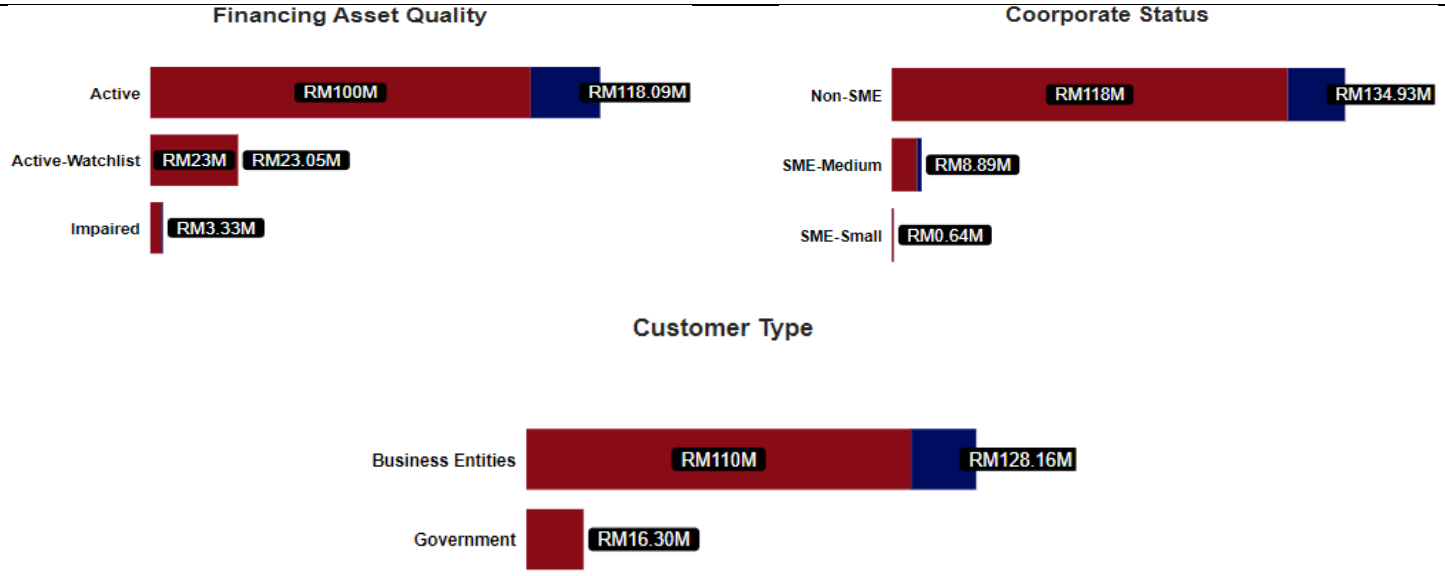


Figure 3.0

Customer Quality:

Even while the cumulative profit payment is decreasing, the financing asset quality movement appears to be better whereby more good customer during September 2023, though it is stable at the time being we should constantly be concerned since the number of our customers (122 to 86) and accounts (166 to 117) is declining, and most of them fall into the Impaired (high provisions) and Watchlist (medium provisions) categories throughout the course of these three years, as shown in Figure 4.0:

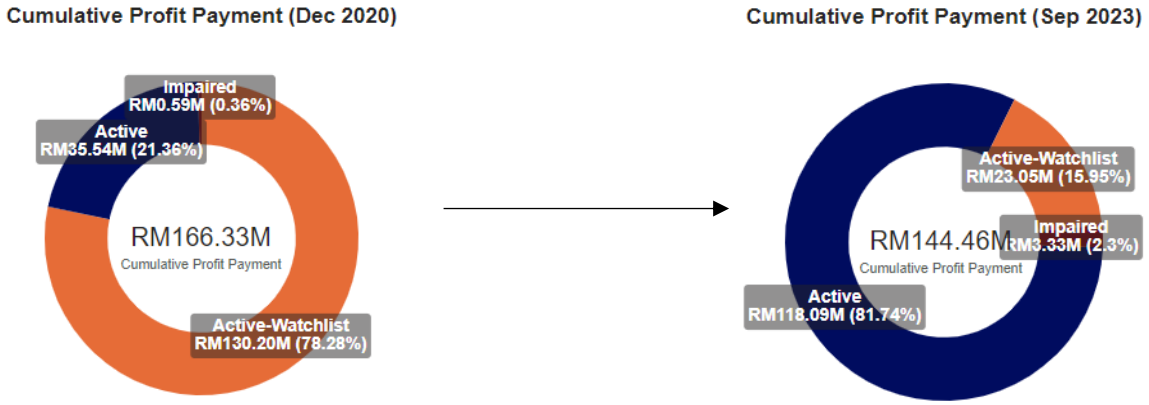


Figure 4.0

Since the decline of customers and accounts in the year 2021, the movement of EXIM Bank customers and accounts reveals a considerable increase for Active customers (37 – 46). As the number of accounts and customers rises, Watchlist and Impaired customer and account numbers eventually declines. There would be less money set aside by the Bank for the year 2023 compared to the year 2021 as a result of lower provisions or costs for Active customers and greater provisions or costs for Watchlist and Impaired customers. The chart below depicts the movement of accounts and customers who are Active, Watchlist, and Impaired:

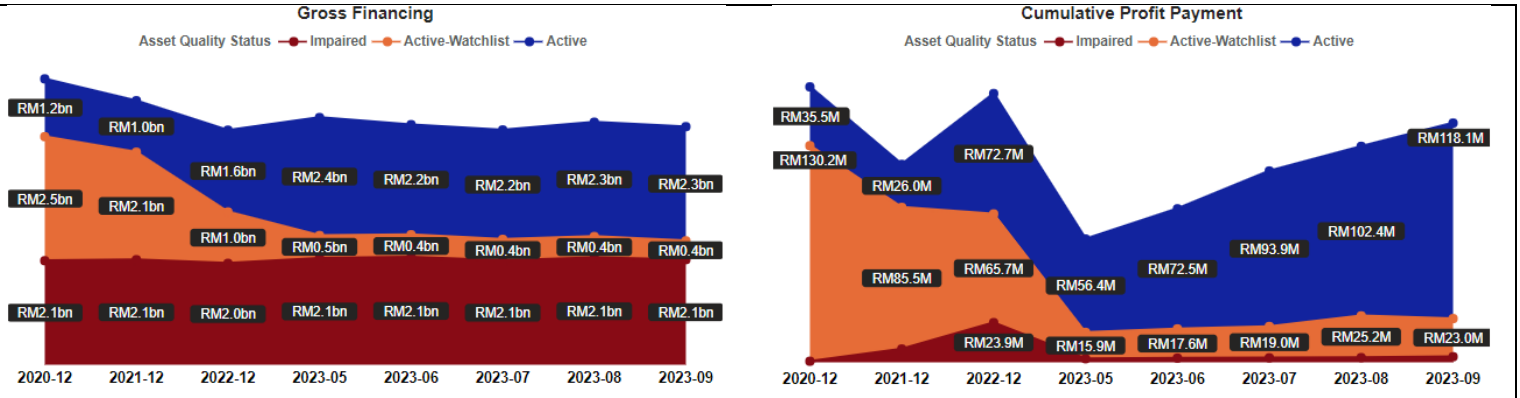


Figure 4.1

Impaired Trend

When the number of accounts for trade and non-trade declines but gross financing assets rise, the trend of impaired accounts becomes alarming. In addition, the Impairment Trend shows a peak from 2018 to 2019 before a sluggish increase in 2021 and 2022:

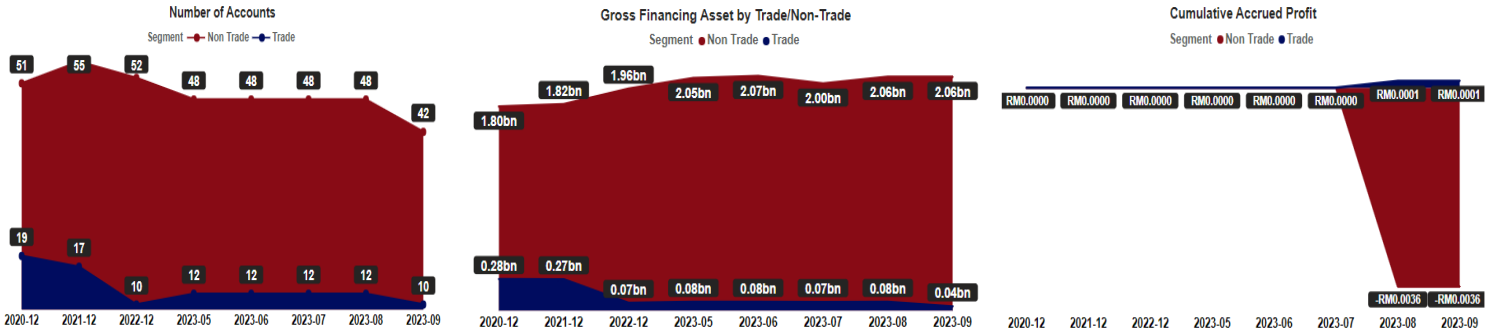


Figure 4.2

Impaired Customer Portfolio: September 2023

Out of 52 number of impaired accounts, there are existing customer still in our book that impaired since 2008. Though we are probably losing our financing asset we are still making profit for some of the accounts as per shown in the figure below:

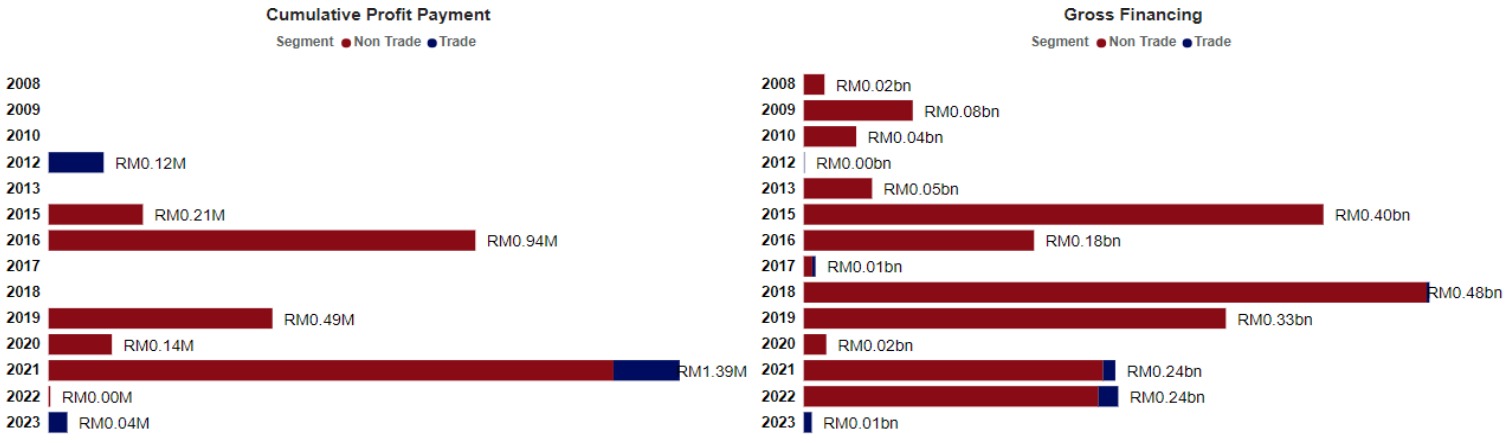


Figure 4.3

Most impaired accounts (18 accounts) are from non-trade financing, buyer credit and overseas project financing account consist of RM 1.30 billion (62%) of the impaired gross **financing asset**. The impaired account portfolio consists of non-MYR currencies and non-syndication/club deals with quarterly payment frequencies. According to the data, the two industries most affected by impairment are mining and transportation, and by country, Indonesia accumulates the longest period of impairment (2008, PT Eden Capital), the most accounts (16), and the highest amount of impairment (RM0.87 billion, 42%).

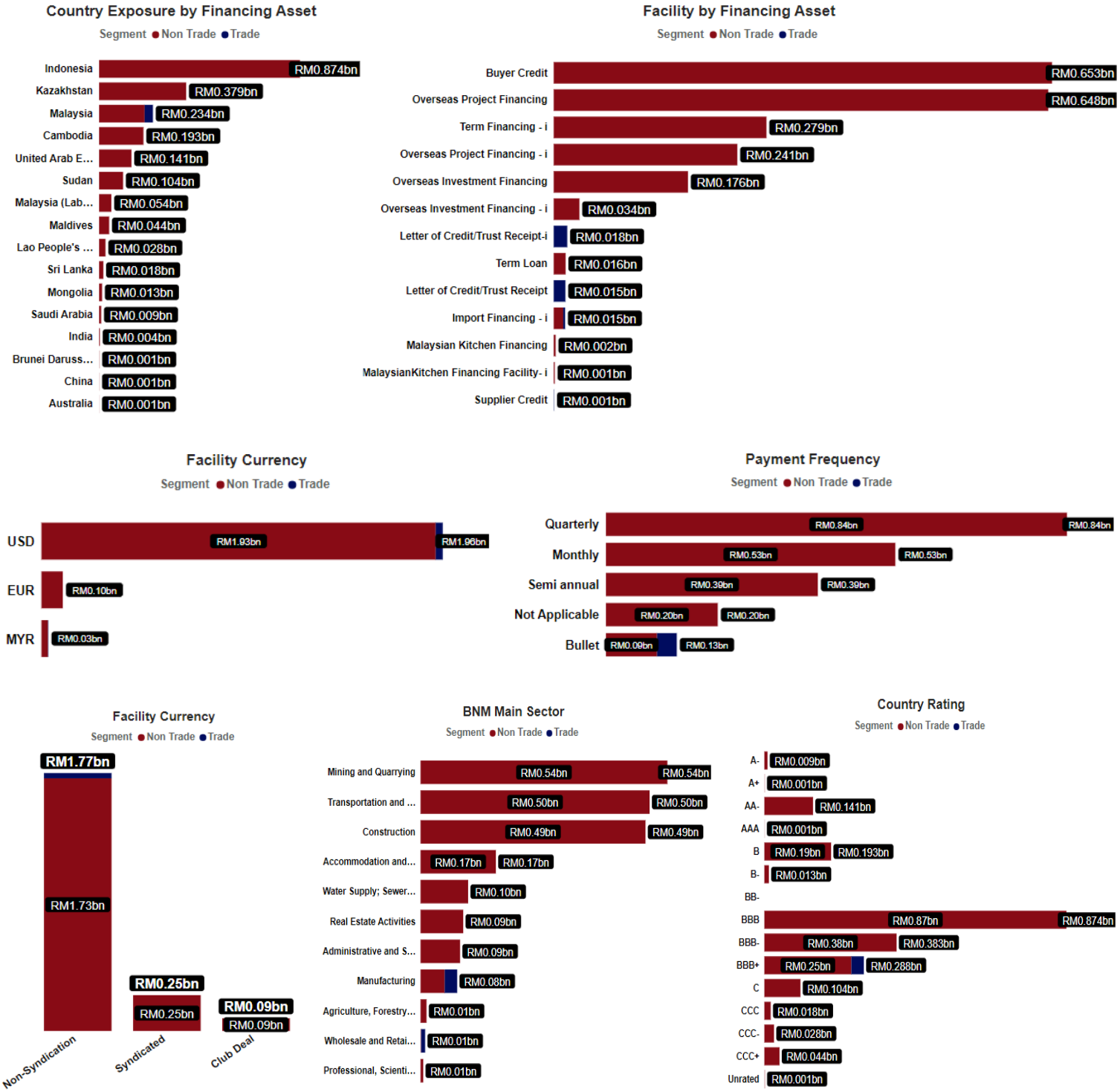


Figure 4.4

Behaviour Prediction

On a normal basis, the average for an account to turn impaired or watchlist is shown in figure 4.5. The interpretation of the figure can be translated by a one-way relationship, If the number of months increases the chances of turning watchlist or impaired decreases. The figures show watchlist has lowest number of months in 2021 while impaired has lowest number of months in 2020. In 2020, EXIM bank customers have a higher chance of becoming impaired after 15 months on board with EXIM, and those who are already impaired have a more difficult time turning watchlist by 46 months, probably due to the phenomenon of the COVID-19 uprising. Same goes for the watchlist. At the end stage of COVID-19, most companies are recovering, and Active customers have an easier chance to turn watchlist by serving 17 months with EXIM Bank but have a difficult time to turn impaired by serving 56 months. In a way, knowing how long it takes for customer churn is a good thing because then it can be monitored and followed up to prevent customer receiving negative experience.

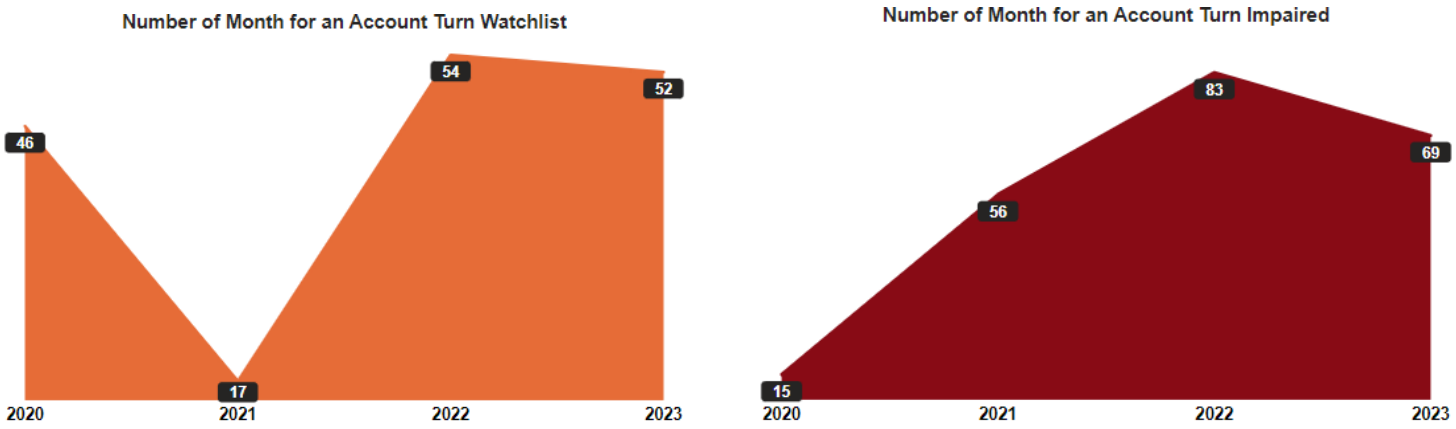


Figure 4.5

Accounts Migration

Table below shows number of accounts by asset quality segmentation throughout the year (Gross Financing and Cumulative Profit Payment can be refer in Figure 4.1):

Financial Asset Quality	Dec 2020	Dec 2021	Dec 2022	Sep 2023
All Accounts	**162**	**138**	**119**	**117**
Active	**37**	**17**	**26**	**46**
Watchlist	**55**	**49**	**31**	**19**
Impaired	**70**	**72**	**62**	**52**

Table 1.0

Table below shows that the number of accounts & profit payment of the new account is slower than the dismissed accounts the number of accounts by migration segmentation (to validate, Ex. 2021 = 162 is (129 + 33) – 138 is (129 + 9) = 24):

All Account				
Date	Dec 2020	Dec 2021	Dec 2022	Sep 2023
Total	**162 Accounts**	**138 Accounts**	**119 Accounts**	**117 Accounts**
Account Movement	**-**	**24 Accounts**	**19 Accounts**	**2 Accounts**
Gross Financing	RM5.70 billion	RM5.27 billion	RM4.68 billion	RM4.75 billion
C. Profit Payment	RM 0.17 billion	RM 0.12 billion	RM 0.16 billion	RM 0.14 billion

Maintain Accounts	-	**129 Accounts**	**110 Accounts**	**101 Accounts**
Gross Financing	-	RM4.81 billion	RM4.32 billion	RM4.49 billion
C. Profit Payment	-	RM0.12 billion	RM0.15 billion	RM0.14 billion

New Accounts	-	**9 Accounts**	**9 Accounts**	**16 Accounts**
Gross Financing	-	RM0.46 billion	RM0.37 billion	RM0.26 billion
C. Profit Payment	-	RM1.73 million	RM7.37 million	RM6.07 million

Dismissed Accounts	-	**33 Accounts**	**28 Accounts**	**18 Accounts**
Gross Financing	-	RM0.437 billion	RM0.802 billion	RM0.249 billion
C. Profit Payment	-	RM13.89 million	RM15.79 million	RM13.47 million
Reason	-	16 – Fully Settled 5 – Cancelled 5 – Conversion 4 – Expired 3 – Fully Write Off	14 – Fully Settled 9 – Fully Write Off 5 – Cancelled	8 – Fully Settled 8 – Fully Write-Off 1 – Cancelled 1 – Conversion

Table 1.1

From the table below, the number of impaired accounts is sluggishly recovering and lesser new account that impaired but most of the dismissed account in a concern when mostly the accounts in fully written off from EXIM book. (to validate, Ex. 2021 = 70 is (61 + 9) – 72 is (61 + 11) = 2):

Impaired				
Date	Dec 2020	Dec 2021	Dec 2022	Sep 2023
Total	**70 Accounts**	**72 Accounts**	**62 Accounts**	**52 Accounts**
Account Movement	**-**	**2 Accounts**	**10 Accounts**	**10 Accounts**
Gross Financing	RM2.08 billion	RM2.10 billion	RM2.02 billion	RM2.10 billion
C. Profit Payment	RM0.59 million	RM7.99 million	RM23.92 million	RM3.33 million

Maintain Accounts	-	**61 Accounts**	**55 Accounts**	**50 Accounts**
Gross Financing	-	RM4.81 billion	RM1.79 billion	RM2.09 billion
C. Profit Payment	-	RM117.77 million	RM18.46 million	RM3.29 million

New Accounts	-	**11 Accounts**	**7 Accounts**	**2 Accounts**
Gross Financing	-	RM0.46 billion	RM0.23 billion	RM0.006 billion
C. Profit Payment	-	RM1.73 million	RM5.47 million	RM0.04 million

Dismissed Accounts	-	**9 Accounts**	**17 Accounts**	**12 Accounts**
Gross Financing	-	RM0.12 billion	RM0.37 billion	RM0.06 billion
C. Profit Payment	-	RM0.40 million	RM0.22 million	RM0.84 million
Reason	-	3 – Fully Write-Off 3 – Fully Settled 3 – Conversion	9 – Fully Write-Off 6 – Fully Settled 2 – Watchlist	8 – Fully Write-Off 3 – Watchlist 1 – Cancelled

Table 1.2

Churn Prediction

The prediction uses logistic regression model, one of supervised machine learning algorithm that is used to predict the probability of an outcome. The model can indicate whether an account will turn impaired or not by setting up the correct parameter to predict those accounts. Listed parameter as per below:

1) Trade/Non-Trade Segmentation
2) Traditional/Non-Traditional Trade Market
3) Customer Payment Frequency
4) Customer Corporate Status
5) Country Region
6) S&P/Moodys Country Rating
7) Customer Main Sector
8) Customer Facility
9) Facility Currency

would like suggest if you want to do this.. please also add random forest as well

The model shows accuracy 81% for active account to turn impaired while 96% for watchlist account to turn impaired. The active model shows there will be a migration of 10 active accounts turns impaired and 10 impaired accounts turn active while the watchlist model indicates there will be a migration of 2 watchlist accounts turns impaired and 7 impaired accounts turn watchlist.

Matrix table:

Active (80.77%)	Suc-Obs	Fail_Obs
Suc-Pred	**42**	10
Fail-Pred	10	**36**

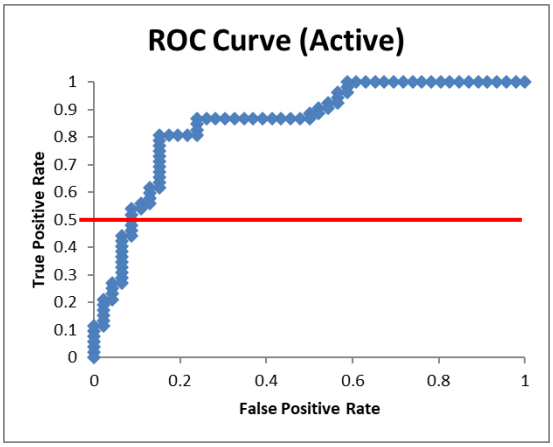
Watchlist (96.15%)	Suc-Obs	Fail_Obs
Suc-Pred	**50**	7
Fail-Pred	2	**12**

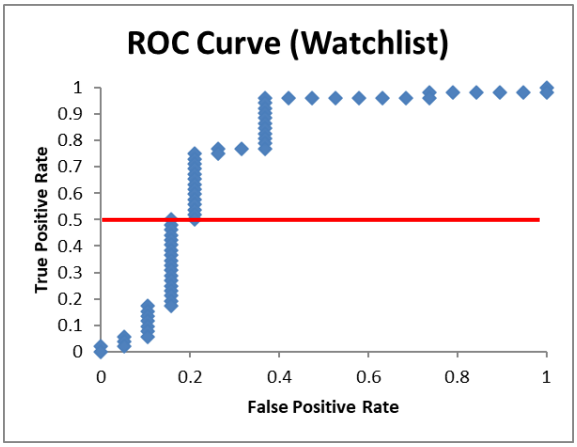
Table 1.3

The Receiver Operating Characteristic (ROC) curves illustrates the diagnostic ability to determine for an account to turn impaired or vice versa. Under the ROC curves the overall account performance based on the 9 parameters listed above, if the probability of impairment for an account is above 0.5 in figure below, we consider for the account as churn to impaired while below 0.5 says otherwise.

i think the word of customer churn is not correct... because here u using impaired.. as far i know customer churn is the percentage of customers who stopped purchasing our business's products or services during a certain period of time

ROC Curve:





are you using ROC as calculate churn rate?

Figure 4.6

Notes: for Impaired Criteria Model 1 (Active) and Impaired Criteria Model 2 (Watchlist) please refer to appendix, figure 6.3 and figure 6.4 respectively.

before coming the calculation of ROC, may i get the result of evaluation metrics w

5. Conclusion

The company has experienced a decrease in total customers, assets, and profits. Most assets and profits are now generated from non-trade activities. As of September 2023, 17% of the company's profits are still to be realized, with 8.8% from trade and 91.2% from non-trade activities. The company's most profitable criteria are those that are non-trade, syndicated, traditional, USD-based, overseas project financing, manufacturing, and non-SMEs in Malaysia where it contradicts with EXIM Bank mandate to provide emphasis on Trade and non-traditional markets.

There is a concern about impaired assets, as 50% of assets do not generate profit. Non-trade activities also contribute to most asset impairments. Indonesia has the most impaired accounts (16 accounts), with the longest impaired account dating back to 2008 (PT Eden Capital Indonesia). The largest impaired asset is also from Indonesia (PT Srivijaya Air).

For all exits accounts, most account are fully settled accounts, which is a good thing as the company does not have to carry losses. However, the company does carry losses for impaired accounts, as some exited accounts are fully written off.

6. Appendix

The following figure lists the top 15 customers in terms of largest loan:

Customer Name	Country Exposure	Asset Quality Status	Facility	Payment Frequency	Loans Outstanding	Accrued Profit	Profit Payment	To Be Paid	Collectable Profit
Yinson Boronia Production B.V.	Brazil	Active	Overseas Project Financing	Quarterly	RM563,800,306.92	RM8,192,434.92	RM32,547,638.08	100.14%	25.17%
PT Sriwijaya Air	Indonesia	Impaired	Buyer Credit	Monthly	RM380,100,012.14	RM0.00	RM0.00	82.50%	0.00%
Markmore Energy (Labuan) Limited	Kazakhstan	Impaired	Overseas Project Financing	Quarterly	RM378,948,389.76	RM0.00	RM0.00	203.23%	0.00%
The Ministry of Finance Government of Lao PDR	Lao People's Democratic Republic	Active-Watchlist	Buyer Credit	Quarterly	RM248,695,532.61	RM2,952,032.82	RM15,699,340.98	75.90%	21.06%
Felda Global Ventures Capital Sdn Bhd	Malaysia	Active	Supplier Financing - i	Bullet	RM240,376,999.32	RM376,949.32	RM8,211,204.09	68.68%	4.59%
OM Materials (Sarawak) Sdn Bhd	Malaysia	Active	Overseas Project Financing	Quarterly	RM226,679,439.39	RM105,596.26	RM14,638,169.08	141.78%	1.44%
S P Setia Berhad	United Kingdom	Active	Term Financing - i	Bullet	RM202,193,262.63	RM831,741.14	RM11,964,285.12	70.44%	6.95%
HCD-PRBSB Co. Ltd.	Cambodia	Impaired	Overseas Project Financing - i	Not Applicable	RM187,208,861.80	RM0.00	RM0.00	95.00%	0.00%
Istanbul Sabiha Gokcen Uluslararasi Havalimani Yatirim Yapim Isletme A.S	Turkiye	Active	Overseas Investment Financing	Semi annual	RM154,741,038.43	RM2,825,334.02	RM4,787,450.98	80.75%	59.02%
Eversendai Corporation Berhad	United Arab Emirates	Impaired	Term Financing - i	Semi annual	RM140,763,666.60	-RM0.00	RM0.00	100.00%	0.00%
INGRESS INDUSTRIAL (MALAYSIA) SDN BHD	Indonesia	Active	Term Financing - i	Monthly	RM131,340,371.95	RM63,013.12	RM5,840,955.34	87.56%	1.08%
Al Manara Water Company Limited	Sudan	Impaired	Buyer Credit	Semi annual	RM104,422,553.53	RM0.00	RM933,163.88	76.73%	0.00%
Impiana Ubud (Labuan) Co. Ltd (formerly known as Impiana Investment Co. Ltd.)	Indonesia	Impaired	Overseas Project Financing	Quarterly	RM103,128,814.05	RM0.00	RM0.00	100.59%	0.00%
Superboom Projects Sdn Bhd	Indonesia	Impaired	Overseas Project Financing	Bullet	RM93,694,364.03	RM0.00	RM0.00	99.84%	0.00%
Hummingbird Energy (L) Inc	Indonesia	Impaired	Overseas Investment Financing	Quarterly	RM89,229,815.47	RM0.00	RM0.00	82.91%	0.00%

Figure 6.0

The following figure lists the top 15 impaired customers in terms of largest loan:

Customer Name	Country Exposure	Facility	Payment Frequency	Loans Outstanding	Accrued Profit	Profit Payment	To Be Paid	Collectable Profit	Year Impaired
PT Sriwijaya Air	Indonesia	Buyer Credit	Monthly	RM380,100,012.14	RM0.00	RM0.00	82.50%	0.00%	2018
Markmore Energy (Labuan) Limited	Kazakhstan	Overseas Project Financing	Quarterly	RM378,948,389.76	RM0.00	RM0.00	203.23%	0.00%	2015
HCD-PRBSB Co. Ltd.	Cambodia	Overseas Project Financing - i	Not Applicable	RM187,208,861.80	RM0.00	RM0.00	95.00%	0.00%	2019
Eversendai Corporation Berhad	United Arab Emirates	Term Financing - i	Semi annual	RM140,763,666.60	-RM0.00	RM0.00	100.00%	0.00%	2022
Al Manara Water Company Limited	Sudan	Buyer Credit	Semi annual	RM104,422,553.53	RM0.00	RM933,163.88	76.73%	0.00%	2016
Impiana Ubud (Labuan) Co. Ltd (formerly known as Impiana Investment Co. Ltd.)	Indonesia	Overseas Project Financing	Quarterly	RM103,128,814.05	RM0.00	RM0.00	100.59%	0.00%	2021
Superboom Projects Sdn Bhd	Indonesia	Overseas Project Financing	Bullet	RM93,694,364.03	RM0.00	RM0.00	99.84%	0.00%	2021
Hummingbird Energy (L) Inc	Indonesia	Overseas Investment Financing	Quarterly	RM89,229,815.47	RM0.00	RM0.00	82.91%	0.00%	2019
Sapura TMC Sdn Bhd	Malaysia	Overseas Investment Financing	Semi annual	RM86,752,965.60	RM0.00	RM3,764.97	99.85%	0.00%	2022
PT Sari Indah Lestari	Indonesia	Buyer Credit	Quarterly	RM84,151,740.74	RM0.00	RM0.00	73.20%	0.00%	2009
Daya Maritime Limited	Malaysia	Term Financing - i	Semi annual	RM56,335,695.07	RM0.00	RM0.00	95.55%	0.00%	2016
Kota Laksamana 101 Limited	Malaysia (Labuan)	Term Financing - i	Monthly	RM53,545,329.20	RM0.00	RM0.00	19.34%	0.00%	2018
Hondaafushi Investment Private Limited	Maldives	Buyer Credit	Quarterly	RM43,805,701.58	RM0.00	RM0.00	53.05%	0.00%	2013
UPL Co. Ltd	Lao People's Democratic Republic	Overseas Project Financing	Monthly	RM27,982,730.28	RM0.00	RM0.00	85.82%	0.00%	2018
Hyrax Oil Sdn Bhd	Sri Lanka	Overseas Project Financing - i	Quarterly	RM18,376,850.14	RM0.00	RM1,241,093.67	67.53%	0.00%	2021

Figure 6.1

The following figure lists the top 15 impaired customers in terms of longest impaired:

Customer Name	Country Exposure	Facility	Payment Frequency	Loans Outstanding	Accrued Profit	Profit Payment	To Be Paid	Collectable Profit	Year Impaired
PT Eden Capital Indonesia	Indonesia	Overseas Project Financing	Quarterly	RM16,022,908.03	RM0.00	RM0.00	85.37%	0.00%	2008
PT Sari Indah Lestari	Indonesia	Buyer Credit	Quarterly	RM84,151,740.74	RM0.00	RM0.00	73.20%	0.00%	2009
PT Bumi International Tankers	Indonesia	Buyer Credit	Quarterly	RM13,428,030.74	RM0.00	RM0.00	12.36%	0.00%	2010
PT Layar Sentosa Shipping Corporation	Indonesia	Buyer Credit	Quarterly	RM13,623,590.22	RM0.00	RM0.00	12.54%	0.00%	2010
PT Mahakarya Inti Buana	Indonesia	Buyer Credit	Quarterly	RM13,487,764.64	RM0.00	RM0.00	11.98%	0.00%	2010
Solartif Sdn Bhd	Malaysia	Supplier Credit	Bullet	RM637,264.90	RM0.00	RM121,014.32	42.48%	0.00%	2012
Hondaafushi Investment Private Limited	Maldives	Buyer Credit	Quarterly	RM43,805,701.58	RM0.00	RM0.00	53.05%	0.00%	2013
Nirmala Villlas Restaurant Sdn Bhd	India	Malaysian Kitchen Financing	Monthly	RM794,049.34	RM0.00	RM0.00	89.62%	0.00%	2013
Virtual Peak Group Sdn Bhd	Malaysia	Overseas Project Financing	Quarterly	RM8,133,918.01	RM0.00	RM0.00	28.89%	0.00%	2013
Eaglexpress Air Charter Sdn Bhd	Saudi Arabia	Term Financing - i	Monthly	RM9,158,961.56	RM0.00	RM207,237.13	26.78%	0.00%	2015
International Halal Meat Exporter LLC	Mongolia	Overseas Project Financing - i	Quarterly	RM13,105,040.71	RM0.00	RM0.00	102.31%	0.00%	2015
Markmore Energy (Labuan) Limited	Kazakhstan	Overseas Project Financing	Quarterly	RM378,948,389.76	RM0.00	RM0.00	203.23%	0.00%	2015
3 Plus 1 Studio Sdn Bhd	China	Malaysian Kitchen Financing	Monthly	RM1,000,848.23	RM0.00	RM0.00	66.72%	0.00%	2016
Al Manara Water Company Limited	Sudan	Buyer Credit	Semi annual	RM104,422,553.53	RM0.00	RM933,163.88	76.73%	0.00%	2016
Daya Maritime Limited	Malaysia	Term Financing - i	Semi annual	RM56,335,695.07	RM0.00	RM0.00	95.55%	0.00%	2016
PT Great Colour Energy	Indonesia	Overseas Project Financing - i	Quarterly	RM16,050,046.28	RM0.00	RM4,500.00	28.75%	0.00%	2016

Figure 6.2

The following table Shows the country rating by its region and country exposure:

Country Rating	Region	Country Exposure
A	Asia	Japan
A-	Middle East	Saudi Arabia
A+	Asia	China
AA-	Asia	Chinese Taipei / Taiwan
		Korea, Republic of (South)
	Europe	United Kingdom
	Middle East	United Arab Emirates
AA+	Americas	Canada
		United States
	Oceania	New Zealand
AAA	Asia	Singapore
	Europe	Germany
	Oceania	Australia
B	Africa	Tanzania, United Republic of
	Asia	Cambodia
	Europe	Turkiye
B-	Africa	Eswatini
		Ghana
	Asia	Mongolia
	Europe	Bosnia and Herzegovina
BB-	Africa	Seychelles
	Americas	Brazil
	Asia	Bangladesh
		Vietnam
BBB		Indonesia
BBB-		India

		Kazakhstan
BBB+		Malaysia
		Malaysia (Labuan)
		Thailand
	Europe	Spain
C	Africa	Sudan
CCC	Asia	Sri Lanka
CCC-	Asia	Lao People's Democratic Republic
CCC+	Asia	Maldives
	Middle East	Iraq
Unrated	Asia	Brunei Darussalam

Table 2.0

Impaired Criteria Model 1 (Active):

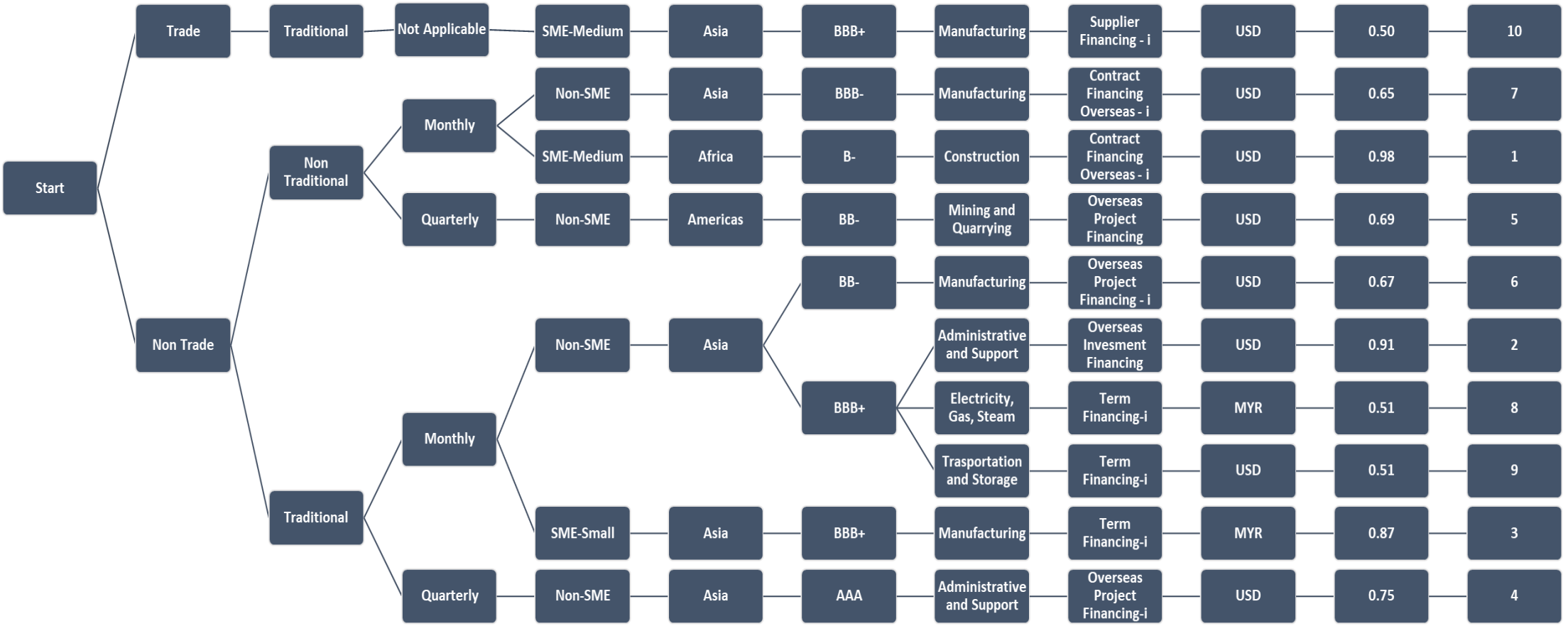


Figure 6.3

Impaired Criteria Model 2 (Watchlist):

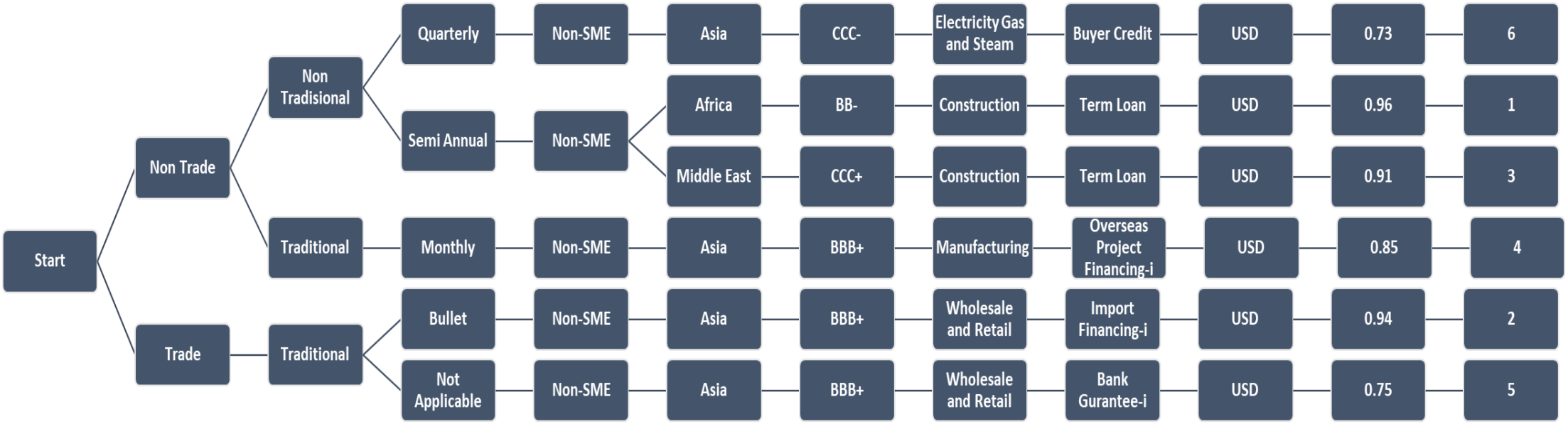


Figure 6.4